IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 27 June 2025, accepted 14 July 2025, date of publication 22 July 2025, date of current version 31 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3591505

RESEARCH ARTICLE

Non-Invasive Glucose Monitoring Using Optical Sensors and Machine Learning: A Predictive Model for Nutritional and Health Assessment

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Universtas Dian Nuswantoro Ethics Committee under Application No. 002351/UNIVERSITAS DIAN NUSWANTORO/2024.

ABSTRACT Glucose monitoring plays a vital role in maintaining metabolic and nutritional balance. However, invasive methods, while accurate, are often painful and impractical for routine use. This study presents a non-invasive glucose monitoring framework that integrates a high-intensity Superbright optical sensor with an IoT-enabled data acquisition system using ESP32 microcontrollers and Raspberry Pi. The IoT-based architecture enables seamless integration with cloud computing platforms, allowing remote access and scalability for large-scale population-level screening and monitoring. The system captures glucose-related optical signals, which are analyzed using various machine learning algorithms, including a novel Convolutional Neural Network–Attention Hybrid Model (CNN-AHM). CNN-AHM combines spatial feature extraction with attention-based prioritization of relevant signal patterns, enhancing both accuracy and interpretability. Among all models tested, CNN-AHM demonstrated the best standalone performance (RMSE = 14.06 mg/dL), representing a 13.18% improvement over its baseline configuration. When combined with complementary strength of the second-best performing Random Forest Regressor (RF-R), the system achieved a substantially improved RMSE of 7.62 mg/dL and an R^2 of 0.83. This work underscores the importance of integrating sensor technology, deep learning, and explainable modeling to advance non-invasive glucose monitoring solutions. Future directions include extending the architecture with explainable AI using SHAP (SHapley Additive Explanations) to provide feature-level interpretability, and exploring real-world deployment scenarios.

INDEX TERMS Convolutional neural network, support vector regression, random forest regression, XGBoost regression, sensor technology, non-invasive glucose monitoring, machine learning in healthcare.

I. INTRODUCTION

Glucose serves as a vital energy source for the body, ensuring optimal physiological and physical functions. Maintaining balanced blood glucose levels is essential for supporting the activity of muscles, the brain, and other critical organs. Conversely, imbalances in glucose levels can lead to significant health concerns. For instance, hyperglycaemia, or elevated blood glucose levels, may indicate risks of diabetes, cardiovascular diseases, and other critical metabolic disorders.

The associate editor coordinating the review of this manuscript and approving it for publication was Patrizia Livreri.

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But if not treated right away, hypoglycemia, or low blood sugar levels, can lead to fainting, seizure frequency, or even a coma [1]. Glucose is very important for children's growth and development because it helps with processes like nervous system and brain development. Children who have problems with their glucose levels, especially during important growth periods, are more likely to have developmental problems and health problems later in life [2]. Therefore, it's very important to keep an eye on glucose levels, especially for groups that are more at risk, like toddlers and elderly people [2], [3]. However, traditional ways of checking glucose levels use invasive methods like blood sampling, which tend to be painful. These problems show how important it is to develop non-invasive glucose monitoring systems that allow for frequent and painless tests. These technologies would make it easier for people to follow their treatment plans and find glucose imbalances early. By keeping a close eye on things, preventative steps and early actions can be taken to keep a healthy diet and overall health. Not only does measuring glucose levels show how carbohydrates are broken down, but it also shows how healthy a person's diet is overall [4], [5]. Studies show that children whose blood sugar levels stay stable tend to be better nourished than those whose levels change a lot [6]. Thus, glucose measurement can provide early information about the risks of nutritional deficiencies or excesses. Frequently monitoring your blood sugar levels is important for figuring out your specific nutritional needs.

Using sensors to monitor glucose levels without invasive method is becoming more popular because they are easy to use and offer little risk to individuals, particularly children [7], [8]. Invasive procedures like taking blood can be frightening and painful, which could make patients less likely to get regular checkups. On the other hand, non-invasive methods make it easier to check on patients more often and without pain, which makes them more comfortable and more likely to follow healthcare rules. Non-invasive methods are also safer for groups that are more likely to get hurt or sick, like toddlers, because they avoid the use of needles.

Advanced technologies for sensors now allow the detection of glucose without piercing the skin by using signals from light or changes in bioimpedance [9], [10] These new technologies that don't require surgery are especially useful in places where resources are limited and access to regular medical tests may be limited. When coupled with machine learning, such technologies support early detection of health risks associated with glucose imbalances, offering a promising alternative to traditional monitoring. In this study, we utilize a high-intensity Superbright optical sensor that emits selective multi-wavelength light to penetrate biological tissues and capture glucose-related optical variations. This sensor technology offers several key advantages, including low cost, energy efficiency, and rapid signal acquisition. By measuring changes in the optical properties of light reflected or absorbed by tissues, superbright sensors can identify patterns related to glucose concentrations [11], [12], [13]. Integrated with machine learning models, the system interprets these sensor signals to provide accurate and real-time glucose level estimations. This approach enhances both diagnostic performance and user experience, paving the way for scalable, non-invasive glucose monitoring in clinical applications.

This research contributes to the growing field of non-invasive health monitoring by introducing an IoT-integrated Superbright sensor system combined with a novel hybrid machine learning architecture. While most traditional glucose monitoring methods remain invasive and uncomfortable for frequent use, our approach offers a user-friendly, scalable, and real-time alternative. The core innovation lies in the development of CNN-AHM, which combines spatial feature extraction with dynamic feature prioritization to enhance prediction performance and interpretability, two critical needs often unmet in prior work. In addressing both technological and clinical challenges, this study offers the following key contributions:

1. IoT-Enabled Superbright Sensor Integration: We present a non-invasive glucose monitoring system that integrates a high-intensity Superbright sensor within an IoT-based framework, enabling real-time data acquisition, processing, and transmission for continuous and remote health monitoring.
2. The integration of CNN with an attention mechanism in a hybrid model: It leverages convolutional layers for feature extraction and an attention mechanism to prioritize critical patterns in the sensor data. This hybrid architecture improves prediction performance while maintaining interpretability, making it particularly suitable for health monitoring systems where understanding model decisions is crucial.
3. To enhance predictive reliability, we strengthen the proposed CNN-AHM model by integrating it with the second-best performing RF-R using an ensemble appraoch, resulting in significantly improved accuracy and robustness for real-world deployment.

II. RELATED WORK

Biosensors represent a transformative technology with the capacity to fundamentally reshape biomedical research and practice. The combination of portability, economic efficiency, and operational simplicity has catalysed substantial market expansion. Diabetes mellitus, a condition characterized by high glucose levels in the body, is one of the most prevalent diseases contributing to mortality worldwide. Although there is no cure or preventive measure, its complications can be managed and mitigated through proper monitoring and treatment. Glucose concentration can be quantified through multiple biological specimens, encompassing blood, sweat, urine, interstitial fluid, tears, respiratory exhalation, and salivary secretions. Numerous innovative techniques to develop biosensors for ongoing blood sugar monitoring have emerged. Researchers have presented a wide range of pioneering techniques to develop biosensors for blood sugar monitoring [14].

Precision nutrition aims to develop tailored dietary guidance by integrating a comprehensive range of individual-specific data, extending beyond traditional diet and physical characteristics to include sophisticated molecular and biological insights such as genetic profiles, gene activity, microbial compositions, protein landscapes, and metabolic signatures. Emerging digital technologies and sophisticated data analysis methodologies, including sensor technologies, machine learning algorithms, and advanced mathematical modelling, have facilitated the development of innovative analytical approaches in personalized nutrition research. This investigation explored the potential of non-invasive vibrational spectroscopic techniques, integrated with advanced data analysis strategies, as promising analytical tools for precision nutrition applications. Specifically, the research examined the comprehensive analysis of metabolites and biomarkers present in alternative biofluids, such as saliva and urine, utilizing infrared and Raman spectroscopic techniques to advance human nutritional understanding [15].

In this study, they addressed the precise estimation of blood sugar concentrations in water using near-infrared (NIR) spectroscopy, which often presents indistinguishable signals. This research comprehensively evaluated multiple machine learning algorithms for spectroscopic data interpretation, systematically exploring and characterizing the distinctive feature extraction capabilities of diverse predictive modelling approaches. The computer methods that were looked into included a variety of regression techniques, including Partial Least Squares Regression, SV-R, RF-R, Extra Tree Regression, Extreme Gradient Boosting (XGBoost), and a hybrid approach that combined Principal Component Analysis (PCA) with Neural Networks. The study's main goal was to find and critically examine the unique ways that each computational model learns features. It calculated 564 NIR absorption spectrum samples in water with glucose levels between 40 and 500 mg/dl. The wavelengths ranged from 900 to 2200 nm. Then, these spectral instances were split into two groups: 80% for training as well as 20% for testing. The evaluation of the machine learning models showed that they were very good at making predictions. SV-R, Extra Tree Regression, and PCA Neural Network all had excellent performance metrics. These models were accurate because their correlation coefficients were higher than 0.99 and their determination coefficients were higher than 0.985. A full feature analysis was done to test the robustness of the computational methods. This involved extracting and comparing high-weight features from different datasets. The study found that the significant impact features determined by algorithms trained on different datasets were very similar, which supports the idea that computational methods are stable and can be used in many different situations. The analysis also pointed out important feature areas that are necessary for separating signals that are hard to tell apart. The study showed that near-infrared spectroscopy could be used to make strong machine learning models that could predict how much glucose was in water in a lab setting [16].

The last few decades have witnessed the continuous development of continuous glucose assessment platforms eliminating invasive procedures that measure blood glucose levels accurately. Traditional fingerstick testing methods, though known for their accuracy, are impractical for frequent daily use due to associated discomfort and the high cost of test strips. While continuous glucose monitoring (CGM) systems, including minimally invasive and non-invasive options, offer an alternative, they remain costly and still necessitate periodic fingerstick calibration. With the rising prevalence of diabetes in low- and middle-income countries, there is an urgent demand for affordable and user-friendly non-invasive glucose monitoring solutions to enhance accessibility and management of the condition. This review briefly discussed non-invasive glucose measurement technologies and related research. The technologies explored encompass optical, transdermal, and enzymatic methods, with a specific focus on NIR technology and NIR Photoplethysmography (NIR PPG) for blood glucose level prediction. The discussion highlighted feature extraction from PPG signals and the application of machine learning techniques for glucose prediction. The review concluded by presenting critical insights and recommendations to guide the future development of blood glucose monitoring systems based on NIR PPG technology [8].

Diabetes is a chronic condition in which the body fails to break down blood sugar effectively due to a lack of insulin. The most critical aspect of healthcare management is maintaining blood glucose levels continuously. A primary concern with effective glucose monitoring devices lies in the reliance on invasive blood-prick techniques. However, these methods may not be suitable for frequent glucose measurements. The paper presented various glucose measurement technologies. It explored non-invasive glucose measurement techniques and their management using advanced medical technologies. Proper configuration of measuring devices is crucial to meet the requirements for non-invasive blood glucose monitoring systems. Non-invasive glucose monitoring devices address the issue of frequent blood pricks for clinical testing. For continuous healthcare monitoring, a Smart Healthcare framework was proposed, integrating the Internet-of-Medical-Things (IoMT) and the Healthcare Cyber-Physical System (H-CPS) for estimating blood glucose levels. The study also discussed several consumer devices and alternative methods for glucose measurement [17].

Nutritional balance is vital for human body functions to avoid adverse health outcomes. Regarding blood analysis, the majority of the literature often overlooks micronutrient balance, which requires more precise regulation. Currently, micronutrient status is routinely assessed through complex methods that are challenging for both patients and physicians. To address the global burden of micronutrient malnutrition, innovations in assessment methods must be accessible and non-invasive. Supporting this objective, this article synthesizes background information on micronutrients, reviews the state of biofluid and physiological analysis for their assessment, and presents actionable opportunities to advance this

field. Adopting a unique clinical perspective absent from technological research on this topic, They found that current practices have limited clinical relevance, a lack of overlap between biofluid and physiological approaches, and highly invasive and inaccessible solutions. Their future work has the opportunity to maximize the impact of new assessment methods by combining clinical relevance, the holistic nature of micronutrients, and prioritizing accessible and non-invasive systems [18].

III. THEORETICAL FOUNDATION

A. PRINCIPLE OF NON-INVASIVE SENSING TECHNOLOGY

Non-invasive sensing technology enables the measurement of biological parameters without penetrating the skin or extracting blood samples, making it safer and more convenient, particularly for vulnerable populations such as children and elderly peoples. This technology relies on detecting physiological signals or biomarker from the body's surface, employing methods such as optical spectroscopy, bioimpedance, and thermal imaging [17]. For instance, optical spectroscopy involves directing light of specific wavelengths onto the skin and analysing the light that is absorbed, reflected, or scattered to detect various biomarkers [15]. NIR and Raman spectroscopy are widely used techniques that facilitate the identification of glucose and other biomolecules based on their unique spectral signatures. Bioimpedance analysis also looks at how well body tissues resist electrical currents, which can tell us about the composition of the body and hydration levels [10]. These non-invasive techniques minimize patient discomfort, reduce the risk of infection, and enable continuous or frequent monitoring, thereby improving compliance and accessibility to health monitoring. Recent advancements in sensor technology and machine learning have significantly enhanced the sensitivity and accuracy of non-invasive methods, making them increasingly viable alternatives to traditional invasive measurements [11], [17], [18]. Consequently, non-invasive sensing technologies hold substantial potential in healthcare for early disease detection, routine monitoring, and personalized health management. Building on the literature regarding the improved capabilities of non-invasive methods and machine learning in health condition assessment, this study explores the use of superbright sensors to predict glucose levels.

Current state-of-the-art non-invasive glucose monitoring systems, such as those employing NIR spectroscopy and Raman spectroscopy, have made significant strides in reducing patient discomfort compared to invasive techniques. However, these methods face several limitations that our proposed approach effectively addresses:

1. Accuracy and reliability: Skin characteristics, levels of hydration, and outside factors can often interfere with NIR and Raman spectroscopy, which can make them less accurate [19], [20]. In contrast, our method combines the Superbright optical sensor along with the CNN-AHM model. It uses attention mechanisms and convolutional layers to uncover significant features, which makes predictions more accurate even when the data is noisy.
2. Cost-effectiveness: NIR and Raman need special, expensive equipment, which makes them hard to get in places with limited resources. The Superbright sensor is a cheaper option that works just as well as the other one but doesn't need expensive hardware, which makes it more likely to be widely used.
3. Ease of use and scalability: Spectroscopy-based systems that are already in use need to be carefully aligned or calibrated, which can make them less useful for everyday tasks. Our IoT-enabled framework uses the Superbright sensor to monitor glucose levels in real time, which makes it easy to collect and process data. This system can be used on the go, is easy to use, and can be expanded, so it can be added to wearable devices or health monitoring systems at home.

Therefore, the proposed system solves some of the biggest problems with present non-invasive glucose monitoring techniques by making them much more accurate, affordable, easy to use, and scalable. By getting around these problems, our method shows a lot of promise for moving the area of glucose monitoring forward, making it a useful and easy-to-use option for many different groups of people.

B. SUPERBRIGHT SENSOR TECHNOLOGIES

The superbright sensor system is a way to keep an eye on the health without having to touch. It uses high-intensity light at different wavelengths to pick up biomedical data from the human body [21], [22]. The sensor sends certain wavelengths of light to the skin, which allows it to go deeper into tissues and interact with molecules like glucose. This means that blood samples are no longer needed. The Superbright sensor detects changes in the absorption of light, reflection, or scattering, which are linked to the amount of certain biomolecules concentration [21]. Then, this method is used to look at these optical signals and turn the raw data into useful health-related metrics for each person. This technology captures data in real time, which makes it perfect for continuous glucose monitoring, particularly among groups of people where traditional invasive methods like blood sampling may not be possible or desirable. The superbright sensor can also be utilized to find other biomarkers besides glucose, which could be useful for broader evaluations of nutrition as well as metabolism [23]. Integrating machine learning with Superbright sensor technology can make it even better at predicting things, which will make health monitoring more accurate. So, the Superbright sensor looks like a good tool for non-invasive diagnostics because it is easy to use and very sensitive and accurate for both personal and clinical health care.

C. MACHINE LEARNING FOR SPECTRAL DATA ANALYSIS

Machine learning is very important for analyzing spectral data because it lets us find and measure biomolecules through complex signal patterns [24], [25]. In non-invasive sensing, spectral information collected by sensors including the

superbright sensor can frequently be high-dimensional and may have noise, consequently they need special analytical methods. There are many ways to decrease dimensionality by only looking at the most important features and getting rid of the rest of the information. Data augmentation methods like Gaussian Noise, which adds statistical variations derived from normal distributions, make machine learning models stronger. Jittering is also a flexible way to add to something that works well in the real world. Then, supervised learning models like Random Forest and neural networks can be trained on labeled spectral data to make accurate predictions about biomarker concentrations [26]. NN networks and other advanced neural network architectures are especially good at finding patterns in spectral data because they can pick up on both spatial and sequential data derived from light signals [27]. Moreover, these models get better over time as more data is collected, which lets them adapt and improve their accuracy. Clustering and other unsupervised learning methods are also useful for finding hidden patterns via spectral data that can help with more research or model development.

Machine learning makes spectral analysis of data from sensors more than just simple detection; it can monitor different health metrics in real time and on a personal level. In general, machine learning gives us the tools we need to turn raw spectral data into useful information, which helps non-invasive monitoring of health technology move forward.

IV. MATERIAL AND METHOD

This chapter explains the methodical approach taken in this study to ensure the reproducibility and accuracy of the findings. It provides detail about the tools, technologies, and methods used, with a focus on non-invasive sensing, glucose monitoring, and machine learning. The choice of sensors, the design of the experiments, and the methods used to prepare the data are all explained in detail to show how they help meet the research goals. Furthermore, the research elaborates on machine learning algorithm implementation and spectral data analysis integration to demonstrate their effectiveness in micronutrient quantification within blood samples.

A. DATASET DESCRIPTION AND PARTICIPANT DEMOGRAPHICS

This study employed purposive sampling to recruit participants willing to undergo both invasive (blood sampling) and non-invasive (Superbright sensor) glucose measurement procedures. A total of 540 samples were collected, ensuring sufficient data for evaluating the reliability of the non-invasive method against standard glucose measurements. The dataset comprises measurements from a gender distribution of 53% male and 47% female, and with BMI values ranging from 15.5 to 38.9 kg/m^2, covering categories from underweight to obesity. This diversity enhances model robustness and prevents biases toward specific subgroups.

To prevent class imbalances during model training, Synthetic Minority Oversampling Technique (SMOTE) [28] was applied, ensuring proportional representation across categories (underweight, normal, overweight). This method helped mitigate overrepresentation, improving generalization. Statistical tests confirmed that demographic factors did not introduce confounding effects in glucose prediction. From the samples, we observes six potential predictive features: systolic blood pressure (systole), diastolic blood pressure (diastole), body mass index (BMI), height, weight and gender, finding significant correlation with manual glucose level as the target attribute using Pearson's correlation (r) formulated as follow:

$$r = \frac{\sum (x_i - \bar{x})^2 (y_i - \bar{y})^2}{\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}} \tag{1}$$

where:

- x_i and y_i: data points for variables x and y.
- $\bar{x}$ and $\bar{y}$: average of the variables x and y.
- $\sum$: summing across all data points.

In the context of Pearson's correlation, p-value assesses the statistical significance of the observed correlation coefficient (r) [29]. It quantifies the probability of observing the given correlation. The p-value for Pearson's correlation is calculated using t-statistic as follows:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \tag{2}$$

where:

- r : Pearson's correlation coefficient.
- n : Number of data points (sample size).
- t: t-statistic, a t-distribution.

TABLE 1. Feature selection using correlation coefficient.

Feature	p-value	Correlation Coefficient (r)
Systole	0.045	0,24
Diastole	0.12	0.05
BMI	0.15	-0.03
Height	0.10	0.02
Weight	0.20	0.01
Gender	0.18	0.00

Table 1 present p-value and r for feature selection. It is a important step as it directly influences the model's performance and generalization. Among those features, it was determined that BMI, diastole and gender did not have a statistically significant relationship with glucose levels ($p > 0.05$). Systole showed a weak but statistically significant positive correlation with glucose levels ($p < 0.05$, $r = 0.24$). Systole showed a weak correlation with glucose levels, and despite its statistical significance, its predictive value was minimal. To ensure model robustness and prevent unnecessary noise, systole was excluded from the final model. The study focuses solely on non-invasive sensor-derived measurements for glucose prediction.

B. ETHICAL CONSIDERATION

Ethical clearance for this research was conducted in compliance with the Helsinki Declaration and approved by the Universitas Dian Nuswantoro Ethics Committee (002351/UNIVERSITAS DIAN NUSWANTORO/2024) due to its involvement of individuals. Verbal consent was obtained from the participating. Participation was voluntary, and participants could withdraw at any time. Participants also were reassured that they could skip any questions/treatment they felt uncomfortable with. In addition, we also ensured that there would be no consequences if participants stopped the procedure at any time because of the feeling of discomfort. All information obtained from the study was kept strictly confidential.

C. IOT-BASED DATA ACQUISITION AND SUPERBRIGHT SENSOR INTEGRATION

The IoT-based data acquisition system employs three key hardware components: (1) the Superbright sensor, (2) the ESP32 microcontroller, and (3) the Raspberry Pi, working synergistically to ensure real-time data collection, preprocessing, and transmission to a cloud platform for comprehensive analysis. The Superbright sensor captures glucose-related optical signals from body tissues, providing raw data for further processing. This data is initially acquired and transmitted wirelessly by the ESP32 microcontroller, designed for efficient data handling and communication. The Raspberry Pi acts as an intermediary, securely transmitting it to the cloud infrastructure. We carefully chose these hardware components because they are very efficient, can be easily expanded, and work well with the IoT framework, which guarantees reliable performance. Figure 1 shows how IoT-based data collection and sensor integration function together.

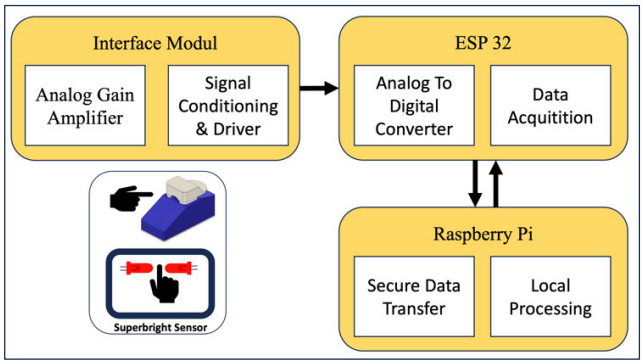

FIGURE 1. The architecture of superbright sensor for non-invasive data acquisition.

Table 2 (in the Appendix section) gives an extensive overview of the functions and requirements of these parts, showing how they work and what their functions are. The Superbright sensor works in the 660–700 nm wavelength range and has an output power of 20–30 mW. Its photodetector is very sensitive and has a noise level below 10–15W/Hz1/2, which reduces interference and makes sure that changes in glucose-related light are detected accurately.

TABLE 2. IoT hardware components.

Components	Functionality	Specification
Superbright Sensor	Capture optical signal for glucose assessment level	Hight-intensity multi-wavelength light
ESP32	Data acquisition and wireless transmission	Wi-fi, Bluetooth, low power consumption
Raspberry-P	Initial preprocessing and securing data transfer	Quad-core processor, 1 GB RAM, 32 GB storage
Power Supply	Uninterrupted power to IoT device	Rechargeable lithium-ion battery (5V, 3,000 mAh)

D. DATASET AND PRE-PROCESSING

We used both invasive as well as non-invasive methods to collect the data for this study. We obtained the non-invasive data from the Superbright sensor, while the invasive data collection involved finger pricking with a lancet to extract blood samples, which were subsequently analyzed using a glucometer. The city's Health Department has checked and confirmed the accuracy of the tools used for invasive measurements. A total of 540 samples were collected, and each sensor measurement gave us about 250 to 350 data points. This two-method approach gives us a solid dataset that lets us directly compare invasive and non-invasive methods to measure glucose level.

The initial step in preprocessing was to make Superbright sensor data easier to analyze. We turned each Superbright column containing comma-separated multiple data points into distinct rows. This transformation put the Superbright data into a standard format, which made a total of 145,436 data points. Data cleaning was then performed to ensures the dataset doesn't have any problems such as missing values, invalid data, duplicates, or null values. Missing values were handled using forward-fill technique, whereas extra missing, invalid data, or unwanted outliers are removed [32]. The dataset has 111,925 data points after it has been cleaned. Furthermore, padding was carried out for making sure that all the array lengths in each column are the same [33]. Once the longest array length was determined, the shorter arrays are then filled with zeros at the end. This makes all the dimensions the same.

Augmentation techniques such as Gaussian Noise and Jittering are incorporated to produce slight variation of data points [30]. It helps prevent overfitting and improves the model's ability to work with different types of data. Jittering is a way to make data more diverse by adding random noise to the original data. This technique makes models more robust to minor variations in the data [30]. The main purpose of jittering is to make the model can recognize patterns even in the presence of slight data variations.

$$x_{\text{jittered}} = x + \text{Noise} \tag{3}$$

where:
- Noise $\sim N\left(0, \sigma^2\right)$
- x_i = source data
- σ = small number to control noise intensity

Gaussian noise values are obtained from a normal distribution with a mean (μ) and standard deviation (σ). This noise is included to mimic the natural changes or disturbances that happen in a lot of physical systems and sensor data. The main purpose is to make models better at generalizing by making them less sensitive to small variations in the data [31]. It also helps the model to avoid overfitting especially when the initial dataset is small or has very regular patterns. The following is the formula for Gaussian noise [32]:

$$x_{\text{noisy}} = x + N\left(\mu, \sigma^2\right) \tag{4}$$

where:
x_{noisy} : data added with Gaussian noise.
x : source data.
$N\left(\mu, \sigma^2\right)$: Noise from normal distribution where μ: mean, σ^2: variants.

After data augmentation, the normalization is done with a MinMaxScaler to make sure that all features are in the same range. This step makes sure that no single feature takes over the learning process because of differences in scale [33]. The data is then split into training and testing sets using an 80%-20% split to obtain evaluation of the model's performance.

E. MACHINE LEARNING ALGORITHMS

This study chose machine learning models based on the need for successfully handling high-dimensional sensor data for non-invasive monitoring of glucose levels. CNN-AHM was selected as the main model because it can use convolutional layers to identify spatial features and an attention mechanism to make the results easier to understand. This makes it perfect for predicting glucose levels in real time. We included SVR, RF-R, XGBoost-R, and k-NNR to evaluate various learning paradigms while making sure that we had a full comparison of performance.

1) SV-R

SV-R is a way for machines to learn for finding the best hyperplane within a certain margin of error [34]. It uses kernel functions, like the Radial Basis Function (RBF), to express non-linear relationships without changing the input space directly [35]. In this study selects SV-R to benchmark performance against other regression models. The following formula [32] is used to compute the prediction:

$$\hat{y} = \sum_{i=1}^{n} \left(\alpha_i - \alpha_i^*\right) K\left(x_i, x\right) + b \tag{5}$$

Where:
- x_i : i^{th} training data.
- x : input data.
- α_i, α_i^* : Lagrange parameters from optimization.
- $K\left(x_i, x\right)$: Kernel function to measure the similarity x_i and x.
- b : Bias.

As shown in Equation 7, the RBF kernel moves the data into a feature space with more dimensions [36], allowing the model to find complex patterns and dependencies that might not be clear in the original feature space.

$$K\left(x_i, x_j\right) = exp\left(-\gamma \parallel x_i - x_j \parallel^2\right) \tag{6}$$

where:
- x_i dan x_j: two vectors in input space.
- $\parallel x_i - x_j \parallel$: distance between x_i and x_j.
- γ : parameter for kernel bandwidth.

2) RF-R

Random Forest Regression (RF-R) is an ensemble learning method that builds multiple decision trees using bootstrapped subsets of the data and averages their predictions to improve accuracy and reduce overfitting. Its ability to handle high-dimensional data and noisy inputs makes it well-suited for biomedical applications [37]. The main steps in RF-R include the following:

Node Split Selection: node splitting is performed to minimize a specific error function, such as Mean Squared Error (MSE). The split is selected to create nodes with the lowest possible error function values.

Prediction by Each Tree: each tree in the Random Forest independently predicts the average target value using the following formula:

$$\hat{y}_{\text{tree}} = \frac{1}{N} \sum_{i=1}^{N} y_i \tag{7}$$

Where:
- N : number of samples in leaf node.
- y_i : sample target value in leaf node.

Prediction Aggregation: The final prediction of the Random Forest Regression is calculated using the following equation:

$$\hat{y}_{\text{final}} = \frac{1}{T} \sum_{t=1}^{T} \hat{y}_{\text{tree}_t} \tag{8}$$

Where:
- T : total number of trees in Random Forest.
- $\hat{y}_{\text{tree}_t}$: prediction made by the tth tree.

Out-of-Bag (OOB) Error: An internal validation metric for Random Forest using the following equation [38]:

$$\text{OOBError} = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{y}_{\text{OOB}}\right)^2 \tag{9}$$

Where:
- y_i : value of actual target.
- $\hat{y}_{\text{OOB}}$: the average prediction of all trees that didn't use i sample during training.

Variance Reduction with Bagging: Random Forest employs the bagging technique to reduce the variance of the model [39] calculated using the following equation:

$$\text{Variance}_{\text{final}} = \frac{\text{Variance}_{\text{tree}}}{T} \tag{10}$$

- $\text{Variance}_{\text{tree}}$: variance prediction from a tree
- T : number of trees in random forest

By increasing the number of trees (T), Random Forest reduces the variance of predictions, thereby enhancing stability and accuracy.

3) XGBOOST-R

XGBoost-R is highly efficient due to its gradient boosting optimization with parallel processing [40], making it faster than other boosting algorithms. Key advantage is its ability to handle irregular data, such as datasets with missing values, as XGBoost-R can automatically manage missing entries. *Prediction Model:* Predictions in XGBoost-R are made by summing the outputs of all the trees formulated as follow:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i) \tag{11}$$

where:

- $f_k(x_i)$: prediction score from the kth tree for data point x_i.
- K : total number of trees in the ensemble model

Objective Function: combines the loss function to measure error and regularization formulated below [41]:

$$O = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{12}$$

where:

- $l(y_i, \hat{y}_i)$: loss function, representing model's error prediction.
- $\Omega(f_k) = \frac{1}{2}\lambda \parallel w_k \parallel^2 + \gamma T_k$: regularization

parameter controlling complexity and overfitting, where λ is weight of regularization parameter, γ is leaf penalization parameter that control the number of leaves in the decision tree, and T_k is leaf number in f_k tree.

Gradient Boosting: new trees are trained using g_i (gradient) and h_i (Hessian) to minimize the loss.

$$g_i = \frac{\partial l(y_i, \hat{y}_i)}{\partial \hat{y}_i}, h_i = \frac{\partial^2 l(y_i, \hat{y}_i)}{\partial \hat{y}_i^2} \tag{13}$$

where:

- g_i : Gradient of the lost function.
- h_i : Hessian of the loss, information about the change of the gradient.

Leaf Node Weight Optimization: The optimal weight for each leaf node (w) is calculated using the following formula:

$$w_j = -\frac{\sum_{i \in I_j} g_i^{(t)}}{\sum_{i \in I_j} h_i^{(t)} + \lambda} \tag{14}$$

- I_j : set of data points that fall into leaf j of the decision tree.
- λ : regularization parameter.

4) k-NNR

k-NNR is a non-parametric algorithm that predicts the target value of a sample by calculating the average target value of its k-nearest neighbors [42]. This algorithm can handle non-linear relationships between features and targets effectively. The distance between sample x_1 and x_1 are calculated using the Manhattan distance, with the following formula:

$$d(x_1, x_2) = \sum_{i=1}^{n} |x_{1i} - x_{2i}| \tag{15}$$

The k-NNR algorithm used in this study is presented in Algorihtm 1 (Appendix section).

```
ALGORITHM 1
Input:
   - Training data X_train = {x1, x2, ..., xn}
   - Target values Y_train = {y1, y2, ..., yn}
   - New sample x_new; Number of neighbors K
   - Distance
Output:
   - Predicted target value y_new for x_new
Algorithm:
  1. Initialize an empty list 'distances'.
  2. For each sample xi in X_train:
     a. Compute the distance between x_new and xi
     b. Append (d, yi) to 'distances' (store the
        distance and corresponding target value).
  3. Sort 'distances' by the first element
     (distance) in ascending order.
  4. Select the first K elements
     Let this subset be 'nearest_neighbors'.
  5. Compute the predicted value y_new:
     a. Initialize sum_y = 0
     b. FOR each neighbour in 'nearest_neighbors':
        sum_y += target value of the neighbour
     c. Compute the average:
        y_new = sum_y/K
  6. Return y_new
```

F. CONVOLUTIONAL NEURAL NETWORK-ATTENTION HYBRID MODEL (CNN-AHM)

CNN-AHM is designed to handle matrix vector data by combining the strengths of CNN and attention mechanisms. It is employed to extract local patterns from sensor matrices, capturing relationships between elements in vectors that reflect critical information. The attention mechanism then highlights more global inter-feature relationships, enabling the model to focus on the most relevant patterns [43]. This combination offers deep analytical capabilities for sensor data, particularly in detecting significant variations. Key components of the model include [44]: *Conv1D and MaxPooling1D Layers:* Conv1D Layers detect local temporal patterns by learning from small groups of features across timesteps, such as identifying trends like increasing values within sequential data. MaxPooling1D Layers reduce data complexity by focusing on the most significant features and improving robustness. *Attention Mechanism:* A dense layer

with softmax activation generates attention weights, assigning varying importance to different features in the sequence. These weights emphasize relevant features to prioritize critical patterns. The flattened feature vector is multiplied by these weights, to focuse on the most meaningful information. *Flatten Layer:* This layer converts multi-dimensional outputs from previous layers- Conv1D and MaxPooling1D, into a one-dimensional vector. The proposed stages are depicted in Figure 2:

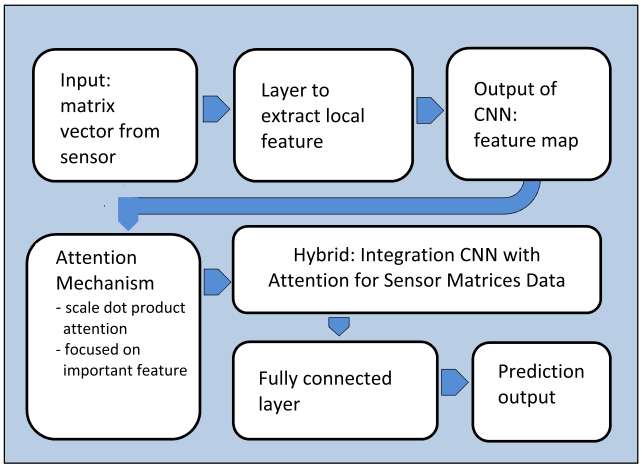

FIGURE 2. The architecture of the proposed CNN-AHM used in this study.

The process that takes place after receiving input data in the form of a matrix from the sensor is described as follows: a. Convolutional Layer – CNN: capturing local relationships within the sensor matrix data. The basic formula for the convolution operation is as follows [44]:

$$f(t) = \sum_{k=1}^{K} w_k \cdot x(t+k) + b \tag{16}$$

where:

- $f(t)$: convolution output at time t.
- K : kernel size, determines the spatial extent of the convolution.
- w_k : weights, define the filter's characteristics.
- $x(t+k)$: input value at time $t+k$.
- b : bias, parameter added to the output.

In this study, max pooling is employed as formulated below [44].

$$p(t) = \max_{t \in \text{window}} f(t) \tag{17}$$

where:

- window: the size of window which specifies the height and width of pooling region.

b. Attention Mechanism: This mechanism is used to assign weights [44] to important sensor data. The scaled dot-product attention formula used attention score as follows:

$$e_t = \text{score}(h_t, s) \tag{18}$$

where:

- h_t : feature representation from CNN at time t
- s : previous state or context vector.

Next, the process continues with Softmax normalization, which transforms the attention scores into probabilities, ensuring that the sum of all weights equals using the following equation:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})} \tag{19}$$

where:

- α_t : attention weight at time t.
- e_t: attention score at time t.

The result from the convolutional layers is then flattened into a one-dimensional vector, serving as the input to the attention mechanism. By amplifying important patterns and minimizing the influence of less significant ones, the attention mechanism increases the interpretability and predictive performance of the model [45]. By combining the capabilities of CNN's feature extraction with the global focus of the attention mechanism, this hybrid architecture gaines a deeper understanding of data, providing more robust predictions.

c. Integration CNN with attention, where the weights are multiplied with the corresponding features. It emphasizes the most significant information in the data using the following equation:

$$c = \sum_t \alpha_t h_t \tag{20}$$

where:

- c : final representation combining all-time-step features with attention weights.

The output from the CNN serves as the input to the Attention Mechanism, where attention assigns weights to the CNN features [46]. The integration of the CNN output and attention is performed as follows:

$$\hat{z} = \text{Attention}(z_{\text{cnn}}, z_{\text{cnn}}, z_{\text{cnn}}) \tag{21}$$

where:

- $\hat{z}$: attention-enhanced feature representation.
- Attention $(\cdot, \cdot, \cdot)$: attention mechanism function with three parameters:(1) Query(Q), the reference point as matching criterion; (2) Key(K), represents the set of elements to be compared against the query Q; (3) Value(V), actual value or feature representations.

This approach describes how CNN(z_{cnn}) features are enhanced through an attention mechanism [47]. The output ($\hat{z}$) contains the amplified combined features.

d. Fully Connected Layer: It generates the final prediction, using the following formula:

$$\hat{y} = \sigma\left(W \cdot \hat{z} + b\right) \tag{22}$$

where:

- $\boldsymbol{\sigma}$: activation function applied to the output of the fully connected layer.

- W : weight matrix of the fully connected layer.
- $\hat{z}$: input vector to the fully connected layer.

During the training of CNN-AHM, scaling and reshaping are consistently applied across both training and testing data to ensure compatibility with the model's architecture. Normalization is performed using a MinMaxScaler, which transforms each feature to a fixed range between 0 and 1. By providing a consistent scale, normalization enhances training stability and improves the model's generalization to unseen data. Reshaping the data into the format (samples, timesteps, features) further ensures compatibility with the 1D CNN layers, allowing them to effectively process sequential data and capture temporal dependencies. For instance, 100 samples, each with 10 timesteps and 5 features, are reshaped to (100, 10, 5).

The CNN-AHM model works well because it incorporates convolutional layers alongside attention mechanisms to find local patterns and dynamically rank the most important features in high-dimensional data from sensors. The convolutional layers read spatial relationships and localized the structure in sequential inputs. The attention mechanism then makes it easier to understand by providing different sensor inputs in different levels of importance. The attention mechanism makes the model focus on prominent patterns without losing useful information, unlike traditional dimensionality reduction methods such as PCA and autoencoders [44]. This hybrid scheme effectively manages complex relationships in the data, keeping spatial and temporal dependencies significant for accurate glucose prediction.

The CNN-AHM model increases computational efficiency and ensures that non-invasive approach works well in real time by using these features. On the other hand, models like k-NNR, RF-R, and SV-R are easier to understand and analyze in terms of feature importance and decision boundaries. However these models have difficulty dealing with sensor data that has a lot of dimensions, especially when it involves capturing temporal dependencies and complex feature interactions. On the other hand, CNN-AHM makes it easier to understand important patterns in the sensor data. This helps to explain how different signal features affect glucose predictions. Deep learning models like CNN-AHM however, need more computing power than RF-R and SV-R. They perform a better task of capturing spatial and sequential dependencies, which makes them more accurate for real-time non-invasive glucose monitoring. We carefully deliberated about this trade-off to make sure that the chosen method perform a good balance between accuracy, model transparency, and practical feasibility.

G. OPTIMIZATION WITH GRID SEARCH

Grid Search is a systematic approach to find the best set of parameters for a model by evaluating every possible configuration in a set parameter space. This method uses cross-validation to check how well the model works with each combination of parameters. This makes it possible to find the best settings that give the best results based on certain evaluation metrics [47]. With this approach, Grid Search ensures that the model achieves maximum performance without relying on manual guesswork for parameter selection. The Grid Search algorithm used in this study is presented in Algorihtm 2 (Appendix section).

```
Algorithm 2
Input:      dataset (X, y), model, parameter_grid,
scoring_metric, cross-validation (cv)
Algorithm:
 1. Split Dataset into Training & Testing
    Sets: X_train, X_test, y_train, y_test ←
    train_test_split(X,y)
 2. Initialize Best Score and Best Parameters:
    a. best_score ← -∞
    b. best_params ← None
 3. For each combination of parameters in
    parameter_grid:
    a. Set model parameters: configure the model
       with the current parameter combination.
    b. Perform cv:
       - Split X_train, y_train into cv folds.
       - For each fold:
         Train the model on (cv_train_X,
         cv_train_y).
         Predict on (cv_valid_X) to get
         predictions.
         Evaluate Performane (MAE, RMSE, MAPE,
         R^2).
       - Compute the average performance score
         across all folds.
    c. Update Best Score and Parameters:
       - If the current score > best_score:
         best_score ← current score.
         best_params ← current parameter
         combination.
 4.  Train the Final Model with Best Parameters:
    a. Configure the model with best_params.
    b. Train the model on the entire training set
       (X_train, y_train).
 5. Evaluate Final Model on Test Set:
    a. Predict y_pred ← model(X_test).
    b. Compute performance for y_test vs. y_pred.
 6. Output:
     - Best Parameters (best_params)
     - Performance Metrics (MAE, RMSE, MAPE, R^2)
```

H. WEIGHTED ENSEMBLE METHOD

Ensemble method is machine learning paradigm where multiple predictive models are combined to achieve superior performance compared to any single model alone [48]. For this study, we adopt a fixed-weight ensemble regression method that combines the outputs of two distinct models which denoted as the top-performing model and the second-best model. This strategy leverages complementary learning strengths by fusing predictions in a weighted fashion. The method is computed as follows:

$$\hat{y}_{\text{ensemble}} = w \cdot \hat{y}_a + (1 - w) \cdot \hat{y}_b \tag{23}$$

Where:

- $\hat{y}_a$: predicted output from the best-performing model
- $\hat{y}_b$: predicted output from 2nd best-performing model
- $\hat{y}_{\text{ensemble}}$: final ensemble prediction
- $w \in [0, 1]$: ensemble weight assigned to Model A

In this workflow, we defer the selection of the best-performing models until post-evaluation, after computing performance metrics (MAE, RMSE, MAPE, R^2). The weights w are set via grid search for balancing emphasis across models.

I. PERFORMANCE EVALUATION OF ALGORITHMS

In this study, the invasive measurements serve as the gold standard for evaluating the performance of non-invasive predictions. The performance of the algorithm is evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2), explained as follows:

1) MAE

MAE measures the average absolute difference between the actual values (y) and the predicted values $\hat{y}$. MAE provides a straightforward understanding of the average error magnitude without considering the direction (positive or negative). This metric is easy to interpret as it has the same unit as the original target. The formula for MAE is as follows [49]:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} | y_i - \hat{y}_i| \tag{24}$$

2) RMSE

The regression metric RMSE calculates the average squared error between the actual and predicted values, then takes the square root. RMSE is more sensitive to large errors compared to MAE, as it assigns greater weight to larger errors through squaring. This metric is often used when it is important to emphasize minimizing significant errors [49].

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2} \tag{25}$$

3) MAPE

MAPE determines the percentage error of the actual values. It provides a more intuitive interpretation in terms of model performance. It computes the relative contribution of error to the target values. However, this metric is sensitive to target values close to zero, therefore it must take the data characteristics into consideration [50].

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^{n} \left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{26}$$

4) COEFFICIENT OF DETERMINATION-R^2

The Coefficient of Determination (R^2) is a metric that calculates how well the model handles the variance in the target data. R^2 values range from 0 to 1, with 1 indicating perfect predictions. This metric is useful for evaluating the proportion of target variability successfully explained by the model compared to a simple mean. R^2 i s often used to assess the overall performance of a model in capturing patterns within the data [51].

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n} (y_i -)^2} \tag{27}$$

TABLE 3. Optimum hyper parameter of the algorithms.

	Hyper parameter	Description	Value
SV-R	C	regularization parameter	00.01
	epsilon	margin of tolerance around predictions.	0.23
	kernel	take data as input and transform it into the required form of processing data	rbf
	gamma	kernel coefficient for 'rbf'	scale
RF-R	max_depth	maximum depth of each tree	10
	min_samples_split	minimum number of samples required to split a node	10
	min_samples_leaf	minimum number of samples required at a leaf node	4
	max_features	obtaining number of features to consider for splitting	sqrt
XGB-R	learning_rate	step size shrinkage to prevent overfitting	0.02
	n_estimators	number of boosting rounds	200
	max_depth	maximum depth of a tree	6
	subsample	fraction of samples used for training each tree	0.08
	reg_alpha	regularization term on weights	0.00
k-NNR	n_neighbors	Number of neighbors to use.	10
	weights	Weight function for neighbors	uniform
	metric	Distance metric to use	Manhattan
CNN-AHM	filters	Number of convolutional filters.	62(1st layer), 32(2nd Layer)
	kernel_size	Size of the convolution kernel.	3
	pool_size	Size of pooling window for down-sampling.	2
	attention_heads	Number of attention heads in the mechanism.	4
	dropout_rate	Fraction of units dropped for regularization.	0.4
	learning_rate	Learning rate for optimization.	0.0003

V. RESULT AND DISCUSSION

A. EXPERIMENTAL SETUP

This study on the use of superbright sensors for glucose prediction employs four machine learning algorithms: SV-R, RF-R, XGBoost-R, k-NNR, and CNN-AHM. Table 3 presents the hyperparameters used for the algorithms after performing Grid Search (Appendix section).

B. THE RESULT

Performance metrics result show varying levels of enhancement, with some algorithms like CNN-AHM demonstrating

significant gains in accuracy and interpretability. This comparison provides valuable insights into the effectiveness of parameter tuning for boosting model performance, making it a crucial step in developing robust predictive models. Table 4 presents the percentage improvement of various algorithm after parameters tuning.

TABLE 4. Performance improvement of algorithms.

	Metrics	With Default parameter	Optimized parameter	% improvement	Normalized MAE & RMSE
SV-R	MAE	11.09	10.88	1.91%	0.1035
	RMSE	15.65	15.65	0.00%	0.1489
	MAPE	10.42%	10.30%	1.15%	-
	R^2	0.15	0.18	18.58%	-
RF-R	MAE	10.16	10.16	0.00%	0.0967
	RMSE	15.65	14.54	7.05%	0.1383
	MAPE	9.67%	9.67%	0.00%	-
	R^2	0.27	0.27	0.00%	-
XGBoost-R	MAE	10.23	10.16	0.78%	0.0967
	RMSE	14.55	14.54	0.05%	0.1383
	MAPE	9.75%	9.67%	0.82%	-
	R^2	0.23	0.27	18.00%	-
k-NNR	MAE	11.22	10.54	6.04%	0.1003
	RMSE	14.55	15.08	3.64%	0.1435
	MAPE	10.73%	10.05%	6.34%	-
	R^2	0.13	0.21	59.53%	-
CNN-AHM	MAE	13.43	11.036	17.80%	0.1050
	RMSE	16.19	14.058	13.18%	0.1338
	MAPE	13.42%	11.29%	15.86%	-
	R^2	0.31	0.479	55.01%	-

We calculated MAE (mg/dL), MAPE (%), RMSE (mg/dL) and R^2 to evaluate the performance of the non-invasive predictions against the invasive gold-standard measurements. These metrics quantify the error and alignment between the non-invasive approach and the reference invasive method. Regarding fairness metrics [52], mean-based normalization presented in 6th column of Table 4 ensures a scale-independent comparison of RMSE and MAE by dividing each metric by the mean of actual glucose values. Only RMSE and MAE are normalized because they are absolute error metrics that depend on the scale of the target variable, whereas MAPE is already percentage-based and R^2 is inherently normalized, making further adjustments unnecessary.

To assess model fairness, we also conducted a subgroup analysis based on BMI categories (underweight, normal, overweight) by evaluating performance disparities using MAE, RMSE, MAPE, and R^2. The results is presented in Figure 3.

The BMI fairness analysis demonstrates that the model maintains stable performance across BMI categories, with MAE (10.75-12.69), RMSE (16.47-18.52), and MAPE (10.56%-12.24%). Compared to CNN-AHM optimized performance (MAE=11.036, RMSE=14.058), the variations across BMI groups remain within an acceptable margin, with a slight increase in error for the underweight category. This indicates that BMI does not significantly impact the model's

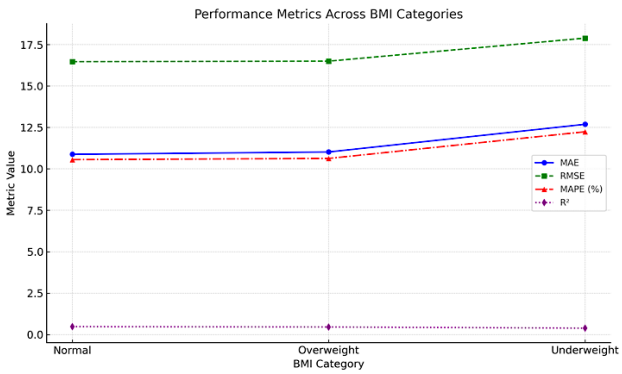

FIGURE 3. Fairness analysis across BMI category.

predictive capability, ensuring equitable performance across diverse physiological conditions. The consistency in MAPE values further confirms the robustness of the model for real-world non-invasive glucose monitoring.

The study also includes a Bland-Altman analysis, a robust statistical method for evaluating agreement between two measurement techniques [53], in this case, invasive and non-invasive glucose monitoring systems. It provides numerical representation of the mean bias and the 95% limits of agreement (LoA), which reflect the degree of agreement and the extent of variability between the methods. In this study, the Bland-Altman analysis demonstrated the following: (1) The calculated mean bias is -1.24 mg/dL in the Bland-Altman analysis indicates that, on average, the model slightly underestimates glucose levels compared to invasive measurements. This underscores that the non-invasive system provides measurements nearly identical to the invasive reference. (2) The 95% Limits of Agreement (LoA) range is [-10.69, 8.21], reflecting a high level of consistency and agreement with only minimal variation between the two methods. This narrow LoA supports the reliability of the non-invasive system for glucose monitoring.mTo offer a more comprehensive insight, we present a detailed explanation of each performance metric and algorithmic performance as follows:

1) MAE

MAE measures the average absolute difference between the actual values (y) and the predicted values $\hat{y}$. It provides a clear and intuitive understanding of a model's accuracy, expressed in the same unit as the target variable. Figure 4 shows the comparison of MAE values achieved by different algorithms Figure 4 XGBoost-R achieved the lowest MAE (10.15), followed closely by RF-R (10.16), which performs almost equivalently. In contrast, CNN-AHM exhibits the highest MAE (11.04), indicating that this algorithm requires additional data or parameter enhancements to match the performance of RF-R and XGBoost-R. A closer analysis of CNN-AHM revealed that the model attempted to predict extreme glucose levels, particularly in the upper range (>180

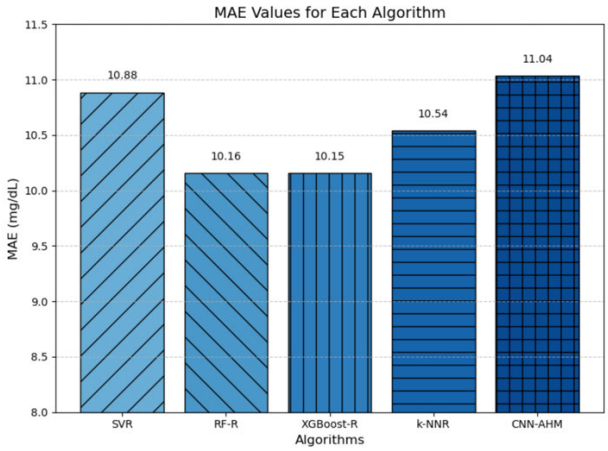

FIGURE 4. MAE performance comparison of individual algorithms.

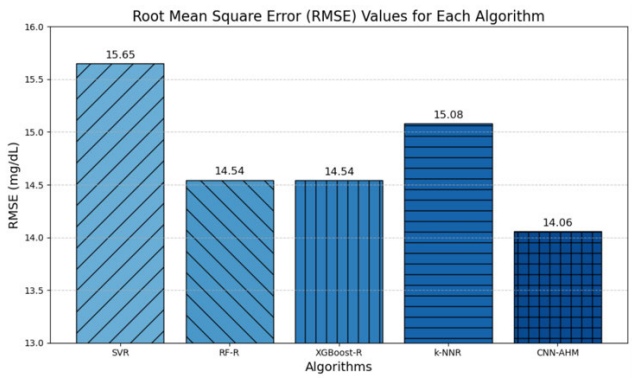

FIGURE 5. RMSE performance comparison of various algorithms.

mg/dL). For example, in one sample with an actual glucose level of 200 mg/dL, the model predicted 184 mg/dL, resulting in a residual error of 16 mg/dL. This discrepancy was due to limited representation of extreme glucose levels in the training dataset. These issues has been mitigated by employing data augmentation techniques to increase the representation of extreme cases or integrating advanced attention mechanisms to more effectively prioritize critical features. Such enhancements have the potential to significantly improve the robustness and accuracy of CNN-AHM in handling challenging scenarios.

2) RMSE

RMSE is a widely used regression metric that calculates the square root of the average squared differences between actual and predicted values.

Unlike MAE, RMSE emphasizes larger errors due to the squaring process, making it highly sensitive to deviations. It provides an effective measure of model accuracy, particularly when minimizing large errors is crucial. Figure 5 shows the comparison of RMSE values achieved by different algorithms. Despite having higher of MAE compared to RF-R and XGBoost-R, CNN-AHM outperforms all other models in RMSE, achieving 14.06. This indicates that CNN-AHM is particularly effective at handling larger errors. On the other hand, both SVR and k-NNR produce higher RMSE values, indicating they are less robust in handling large deviations in the dataset.

3) MAPE

MAPE is a regression metric that measures prediction accuracy by expressing errors as a percentage of the actual values. This makes it particularly useful for understanding the relative magnitude of errors, regardless of the scale of the target variable. However, MAPE can be sensitive to very small actual values, which may inflate percentage errors disproportionately. Figure 6 shows the comparison of MAPE values achieved by different algorithms:

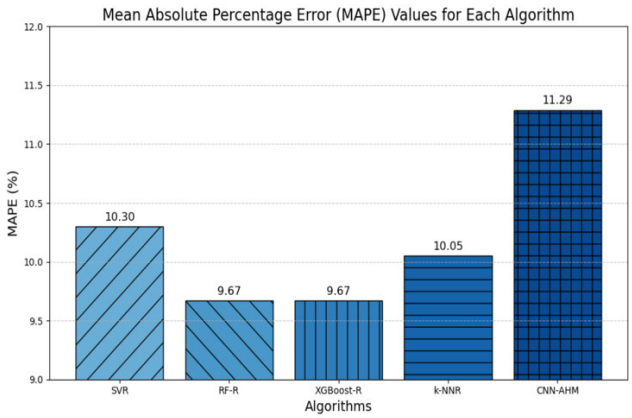

FIGURE 6. MAPE performance comparison of various algorithms.

Both RF-R and XGBoost-R achieve the lowest MAPE at 9.67%, indicating their exceptional accuracy in predicting relative to the actual values. Their robust ensemble approaches help reduce errors consistently across a wide range of values. While SVR (10.30%) and k-NNR (10.05%) perform relatively well, they fall short of RF-R and XGBoost-R, likely due to their lower robustness against variability in the dataset

4) R^2 (COEFFICIENT OF DETERMINATION)

R^2, or the Coefficient of Determination, measures how well a regression model explains the variance in the target variable. It ranges from 0 to 1, where a value closer to 1 indicates that the model explains a large proportion of the variance. R^2 is used for evaluating the overall performance of regression models, with higher values indicating better model fit. Figure 7 shows the comparison of R^2 values achieved by algorithms:

5) CONCLUSION BASED ON VOTING

To determine the best algorithm based on overall performance across MAE, RMSE, MAPE, and R^2, we assign a voting mechanism where each metric receives one vote, and the algorithm with the best value for a specific metric gets that vote. For MAE, RMSE, and MAPE, lower values indicate

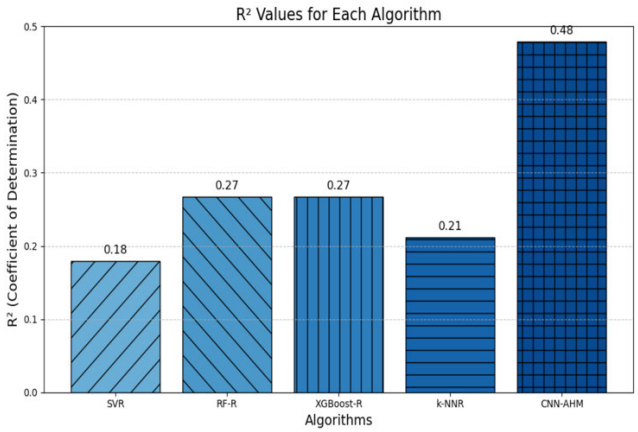

FIGURE 7. R^2 performance comparison of various algorithms.

better model performance, whereas R^2 improves with higher values.

TABLE 5. Voting of algorithm performance.

Metric	Winner	Algorithm Votes
MAE	RF-R	1
RMSE	CNN-AHM	1
MAPE	RF-R & XGBoost-R	2 (shared)
R^2	CNN-AHM	1

Voting as presented in Table 5 is a systematic approach to evaluate and compare the performance of multiple algorithms across different metrics by assigning votes to the best-performing algorithm for each metric. This method ensures a balanced decision-making process, accounting for various aspects of model performance such as accuracy, error minimization, and explained variance. CNN-AHM wins RMSE and R^2, highlighting its strength in capturing variance and minimizing larger prediction errors. However, RF-R emerges as the most consistent algorithm, winning MAE and sharing the MAPE vote with XGBoost-R. It balances low absolute and relative prediction errors, making it a strong candidate choice for this dataset.

6) ENSEMBLE MODEL PERFORMANCE

The proposed hybrid framework achieved promising predictive performance on the full dataset of 540 samples. For deployment setting, we adopted an ensemble strategy that integrates CNN-AHM with the second-best performing RF-R to enhance predictive reliability and mitigate the limitations inherent to standalone models. The ensemble model yielded an MAE of 5.65 mg/dL, RMSE of 7.62 mg/dL, MAPE of 5.42%, and R^2 of 0.83. These results indicate high accuracy and strong explanatory power, validating the model's effectiveness for non-invasive glucose prediction. Figure 8 illustrates the prediction results of the ensemble model, constructed by stacking of CNN-AHM and RF-R, highlighting the model's ability to closely track the actual glucose values across all samples.

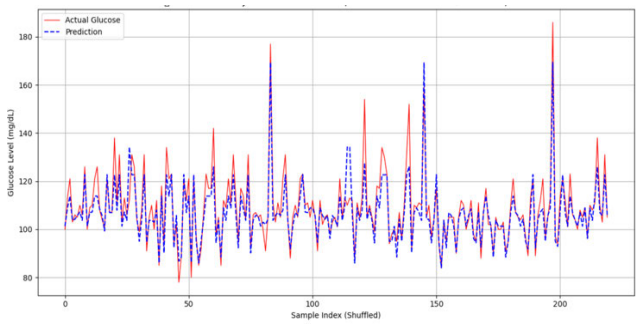

FIGURE 8. Visualization of predicted vs actual glucose values across samples using ensemble method.

Implementing the Superbright sensor for non-invasive glucose monitoring faces challenges despite promising results based on Bland-Altman analysis. Ambient light and temperature changes, as well as other environmental noise, can affect the quality of signals, so advanced denoising algorithms are needed to make sure they are reliable. Calibration drift and differences in skin tone or body composition make it even harder to get consistent measurements. This shows how important it is to have adaptive algorithms and personalized modeling. Also, it is very important to make sure that the sensors are in the right place and that the users follow the rules, because using them wrong can cause big mistakes. Regulatory hurdles and clinical validation require thorough testing on a wide range of people to meet international standards and make sure the product can be used by everyone. Addressing these challenges involves integrating robust signal processing techniques, self-calibration mechanisms, and machine learning enhancements. Even with these problems, the Superbright sensor's ability to provide accurate and real-time glucose monitoring shows a lot of promise for changing non-invasive healthcare solutions.

C. DISCUSSION

The comparative evaluation of the five algorithms reveals notable variations in performance across multiple metrics, as illustrated in Figures 4 to 7. Among all models tested (based on voting as presented in Table 5), CNN-AHM achieved the best standalone performance, followed by RF-R. CNN-AHM showed significant improvements after optimization, achieving an MAE of 11.04, RMSE of 14.06, MAPE of 11.29%, and R^2 of 0.479. These results demostrates the advantage of combining convolutional layers (which effectively capture spatial signal variations) with multi-head attention (which adaptively prioritizes relevant signal patterns). To validate the unique contribution of the hybrid structure, we performed an ablation analysis by removing the attention module, which resulted in a drop in R^2 to 0.31 and increased RMSE to 17.52 mg/dL. This confirms that the attention mechanism significantly enhances the

model's explanatory power and predictive accuracy. To contextualize the performance of our model, we compared the result with recent non-invasive glucose prediction studies. Zeynali et al. [54] reported R^2 values of 0.40 (1-second segment) and 0.29 (10-second segment) using a ResNet34 model, with RMSEs exceeding 27 mg/dL. Our model achieved a higher R^2 and a lower RMSE, demonstrating improved explanatory power and accuracy. Although Naresh et al. [16] achieved an R^2 of 0.99 using in-vitro NIR data, such results are difficult to generalize beyond laboratory settings. Similarly, Sun and Kosmas [55] used a hybrid model combining NIR and mm-wave sensing with metaforests, reporting RMSE and MAE values of 31.83 mg/dL and 17.47 mg/dL, respectively, but did not include R^2, aligning with trends in biomedical modeling where interpretability remains a challenge.

To further enhance predictive reliability and address the limitations of standalone models, we implemented an ensemble strategy by combining CNN-AHM with the second-best performer, RF-R. This integration significantly boosted model performance, resulting in an MAE of 5.65 mg/dL, RMSE of 7.62 mg/dL, MAPE of 5.42%, and a substantially improved R^2 of 0.83. The marked improvement in explanatory power from 0.479 to 0.830 reflects the model's ability to leverage the complementary strengths of both models-CNN-AHM's proficiency in temporal pattern extraction and RF-R's robustness in capturing nonlinear relationships. These findings validate the practical effectiveness of the proposed hybrid architecture in real-world, non-invasive glucose prediction scenarios, addressing both the accuracy and interpretability gaps highlighted in prior studies. Additionally, we incorporated a Bland-Altman analysis, which showed that more than 95% of the predicted glucose levels fall within the acceptable range of agreement. This means that the predicted values are very close to the reference values. We also conducted a Fairness Analysis across BMI groups. The result showed that the model works well across all physiological subgroups. All of these results confirm that the ensemble model for non-invasive glucose prediction works reliable, and can be used in a wide range of real-world situations.

When choosing the right models for application in the real world, computational cost is equally important as predictive performance. RF-R, SV-R, and XGBoost-R are standard machine learning algorithms that are easy to use and require less time or resources to train. These models are suitable for low-power devices with limited processing capabilities. The CNN-AHM hybrid model, on the other hand, is more challenging because it needs more time to train along with additional memory. But the additional effort is beneficial as it finds better high-dimensional and non-linear patterns in spectral sensor data. Once trained, the CNN-AHM enhanced with ensemble method (as part of a hybrid approach) can be used for real-time inference on edge devices with optimized architectures. This makes it possible to use in embedded healthcare systems. Considering the environment where the application will be used and its needs, the trade-off between how hard it is to compute, and how well it can predict should be taken into account.

For future study, several promising directions can be further explored. First, adding advanced attention mechanisms like transformer-based attention mechanism to CNN-AHM could help the model capture more complex dependencies in the data. Second, using domain-specific and advanced augmentation techniques, to focus more on augmentation and feature engineering. It can make the dataset more diverse thereby help generalizing better. Third, adding explainability tools such as SHAP and LIME. It makes models easier to understand and can help user better interpret the model predictions. This case is important in the fields like healthcare where transparency is crucial. Finally, performing scalability studies to see how well the algorithm works on bigger and varied datasets will make sure the robustness and applicability across a broader range of real-world scenarios.

VI. CONCLUSION

This study presents the performance of five machine learning algorithms, i.e., SVR, RF-R, XGBoost-R, k-NNR, and CNN-AHM on dataset to evaluate their predictive accuracy in various metrics, including MAE, RMSE, MAPE, and R^2. RF-R emerged as a reliable model with consistent performance across all metrics, demonstrating its robustness, especially for tabular data. Similarly, XGBoost-R produced highly effective prediction, leveraging gradient boosting to obtain competitive results and highlighting its ability to handle complex patterns and missing data. The k-NNR algorithm, while very intuitive and simple, showed limitations in scalability and struggled with higher-dimensional data. SVR, despite its simplicity, performed well on smaller, well-behaved datasets but was less effective with the complexity of this study's data. The CNN-AHM model, although computationally intensive, showcased significant potential for spectral data and improved its R^2 score after optimization and ensemble method integration. CNN-AHM achieves high accuracy, closely approximating invasive glucose measurements, while preserving the advantages of non-invasive techniques such as reduced discomfort and real-time monitoring. The robustness of the model is reflected in its ability to handle high-dimensional sensor data efficiently, extracting meaningful patterns while mitigating noise. Additionally, the model's reliability is validated through extensive evaluation across varied physiological conditions, including differences in skin tone, tissue composition, and BMI, which can impact optical-based glucose measurements. The model ensures consistent performance across different individuals, reducing bias and enhancing generalizability.

Future work should focus on advancing attention mechanisms, explainability tools, and scalability testing on larger datasets will ensure the continued development of effective and interpretable predictive models. This comprehensive evaluation underscores the importance of tailoring algorithms to specific data characteristics and optimizing them to achieve the best possible performance. In this study, we focused on

hybrid machine learning models to predict glucose levels from non-invasive sensor data. While these models demonstrated significant potential, our future work will explore time-series models, such as time-series Language Model (LLM) [56], Temporal Convolutional Networks (TCN) [57], or attention-based architectures like Transformers [58]. These models are well-suited for capturing temporal dependencies in sequential data and could enhance the performance and interpretability of predictive models. For example, integrating LLMs could allow for a understanding of glucose patterns by leveraging their ability to model long-term dependencies and complex relationships within the data. Future work could also involve fine-tuning pre-trained LLMs specifically for glucose prediction tasks, employing strategies to adapt these models to the characteristics of optical sensor data. This integration could significantly improve the robustness, scalability, and generalizability of non-invasive glucose monitoring systems, paving the way for broader applications in healthcare and nutrition monitoring.

REFERENCES

[1] N. A. ElSayed et al., "6. Glycemic goals and hypoglycemia: Standards of care in diabetes—2024," *Diabetes Care*, vol. 47, no. 1, pp. S111–S125, Dec. 2023, doi: 10.2337/dc24-s006.

[2] E. Rosenfeld and P. S. Thornton. (2023). *Hypoglycemia in Neonates, Infants, and Children*. Endotext. Accessed: Oct. 10, 2024. [Online]. Available: https://www.ncbi.nlm.nih.gov/books/NBK594592/

[3] J. I. Wolfsdorf and C. A. Stanley, "Hypoglycemia in the toddler and child," in *Sperling Pediatric Endocrinology*, 5th ed., Philadelphia, PA, USA: Elsevier, 2021, pp. 904–938, doi: 10.1016/B978-0-323-62520-3.00023-3.

[4] J. A. Wali et al., "Impact of dietary carbohydrate type and protein–carbohydrate interaction on metabolic health," *Nature Metabolism*, vol. 3, no. 6, pp. 810–828, Jun. 2021, doi: 10.1038/s42255-021-00393-9.

[5] C. E. Rauch, A. J. McCubbin, S. K. Gaskell, and R. J. S. Costa, "Feeding tolerance, glucose availability, and whole-body total carbohydrate and fat oxidation in male endurance and ultra-endurance runners in response to prolonged exercise, consuming a habitual mixed macronutrient diet and carbohydrate feeding during exercise," *Frontiers Physiol.*, vol. 12, Jan. 2022, Art. no. 773054, doi: 10.3389/fphys.2021.773054.

[6] C. Piona, M. Marigliano, E. Mozzillo, F. Di Candia, A. Zanfardino, D. Iafusco, G. Maltoni, S. Zucchini, M. Delvecchio, and C. Maffeis, "High glycemic variability is associated with worse continuous glucose monitoring metrics in children and adolescents with type 1 diabetes," *Hormone Res. Paediatrics*, vol. 94, nos. 9–10, pp. 369–373, Dec. 2021, doi: 10.1159/000521430.

[7] S. Laha, A. Rajput, S. S. Laha, and R. Jadhav, "A concise and systematic review on non-invasive glucose monitoring for potential diabetes management," *Biosensors*, vol. 12, no. 11, p. 965, Nov. 2022, doi: 10.3390/bios12110965.

[8] A. Hina and W. Saadeh, "Noninvasive blood glucose monitoring systems using near-infrared technology—A review," *Sensors*, vol. 22, no. 13, p. 4855, Jun. 2022, doi: 10.3390/s22134855.

[9] C. Srichan, W. Srichan, P. Danvirutai, C. Ritsongmuang, A. Sharma, and S. Anutrakulchai, "Non-invasively accuracy enhanced blood glucose sensor using shallow dense neural networks with NIR monitoring and medical features," *Sci. Rep.*, vol. 12, no. 1, p. 1769, Feb. 2022, doi: 10.1038/s41598-022-05570-8.

[10] A. Kandwal, Y. D. Sharma, R. Jasrotia, C. C. Kit, N. Lakshmaiya, M. Sillanpää, L. W. Liu, T. Igbe, A. Kumari, R. Sharma, S. Kumar, and C. Sungoum, "A comprehensive review on electromagnetic wave based non-invasive glucose monitoring in microwave frequencies," *Heliyon*, vol. 10, no. 18, Sep. 2024, Art. no. e37825, doi: 10.1016/j.heliyon.2024.e37825.

[11] Y. Yu, J. Huang, J. Zhu, and S. Liang, "An accurate noninvasive blood glucose measurement system using portable near-infrared spectrometer and transfer learning framework," *IEEE Sensors J.*, vol. 21, no. 3, pp. 3506–3519, Feb. 2021, doi: 10.1109/JSEN.2020.3025826.

[12] J. C. Martínez, V. Vélez-Marín, M. Lopez-Perez, D. F. Patiño-Lugo, and I. D. Florez, "Diagnostic accuracy of the point-of-care standard G6PD test^{TM} (SD biosensor) for glucose-6-phosphate dehydrogenase deficiency: A systematic review and meta-analysis," *Malaria J.*, vol. 23, no. 1, p. 327, Nov. 2024, doi: 10.1186/s12936-024-05144-1.

[13] N. González-Viveros, J. Castro-Ramos, P. Gómez-Gil, H. H. Cerecedo-Núñez, F. Gutiérrez-Delgado, E. Torres-Rasgado, R. Pérez-Fuentes, and J. L. Flores-Guerrero, "Quantification of glycated hemoglobin and glucose in vivo using Raman spectroscopy and artificial neural networks," *Lasers Med. Sci.*, vol. 37, no. 9, pp. 3537–3549, Sep. 2022, doi: 10.1007/s10103-022-03633-w.

[14] V. S. Reddy, B. Agarwal, Z. Ye, C. Zhang, K. Roy, A. Chinnappan, R. J. Narayan, S. Ramakrishna, and R. Ghosh, "Recent advancement in biofluid-based glucose sensors using invasive, minimally invasive, and non-invasive technologies: A review," *Nanomaterials*, vol. 12, no. 7, p. 1082, Mar. 2022, doi: 10.3390/nano12071082.

[15] N. Dongdong and D. Cozzolino, "Possibilities on the application of vibrational spectroscopy and data analytics in precision nutrition," *TrAC Trends Anal. Chem.*, vol. 163, Jun. 2023, Art. no. 117067, doi: 10.1016/j.trac.2023.117067.

[16] M. Naresh, V. S. Nagaraju, S. Kollem, J. Kumar, and S. Peddakrishna, "Non-invasive glucose prediction and classification using NIR technology with machine learning," *Heliyon*, vol. 10, no. 7, Apr. 2024, Art. no. e28720, doi: 10.1016/j.heliyon.2024.e28720.

[17] P. Jain, A. M. Joshi, S. P. Mohanty, and L. Reddy Cenkeramaddi, "Non-invasive glucose measurement technologies: Recent advancements and future challenges," *IEEE Access*, vol. 12, pp. 61907–61936, 2024, doi: 10.1109/ACCESS.2024.3389819.

[18] A. Balch, M. A. Cardei, S. Kranz, and A. Doryab, "Towards an accessible, noninvasive micronutrient status assessment method: A comprehensive review of existing techniques," 2024, *arXiv:2408.11877*.

[19] S. Delbeck and H. M. Heise, "Evaluation of opportunities and limitations of mid-infrared skin spectroscopy for noninvasive blood glucose monitoring," *J. Diabetes Sci. Technol.*, vol. 15, no. 1, pp. 19–27, Jan. 2021, doi: 10.1177/1932296820936224.

[20] B. Gardner, P. Matousek, and N. Stone, "Self-absorption corrected non-invasive transmission Raman spectroscopy (of biological tissue)," *Analyst*, vol. 146, no. 4, pp. 1260–1267, 2021, doi: 10.1039/d0an01940b.

[21] H. Zhou, D. Li, X. Hui, and X. Mu, "Infrared metamaterial for surface-enhanced infrared absorption spectroscopy: Pushing the frontier of ultrasensitive on-chip sensing," *Int. J. Optomechatronics*, vol. 15, no. 1, pp. 97–119, Jan. 2021. [Online]. Available: https://www.tandfonline.com/doi/abs/10.1080/15599612.2021.1953199

[22] J. Lifante, Y. Shen, I. Zabala Gutierrez, I. Rubia-Rodríguez, D. Ortega, N. Fernandez, S. Melle, M. Granado, J. Rubio-Retama, D. Jaque, and E. Ximendes, "Reaching deeper: Absolute in vivo thermal reading of liver by combining superbright Ag_2S nanothermometers and in silico simulations," *Adv. Sci.*, vol. 8, no. 9, May 2021, Art. no. 2003838, doi: 10.1002/advs.202003838.

[23] R. Del Caño, T. Saha, C. Moonla, E. De la Paz, and J. Wang, "Ketone bodies detection: Wearable and mobile sensors for personalized medicine and nutrition," *TrAC Trends Anal. Chem.*, vol. 159, Feb. 2023, Art. no. 116938, doi: 10.1016/j.trac.2023.116938.

[24] A. Chattopadhyay, S. Mishra, and A. González-Briones, "Integration of machine learning and IoT in healthcare domain," in *Hybrid Artificial Intelligence and IoT in Healthcare*. Singapore: Springer, 2021, pp. 223–244, doi: 10.1007/978-981-16-2972-3_11.

[25] Y. Ding, Y. Sun, C. Liu, Q.-Y. Jiang, F. Chen, and Y. Cao, "SERS-based biosensors combined with machine learning for medical application," *ChemistryOpen*, vol. 12, no. 1, Jan. 2023, Art. no. 202200192, doi: 10.1002/open.202200192.

[26] X. Liu, H. An, W. Cai, and X. Shao, "Deep learning in spectral analysis: Modeling and imaging," *TrAC Trends Anal. Chem.*, vol. 172, Mar. 2024, Art. no. 117612, doi: 10.1016/j.trac.2024.117612.

[27] A. Kumar, D. Dembla, S. Tinker, and S. B. Khan, *Handbook of Deep Learning Models for Healthcare Data Processing: Disease Prediction, Analysis, and Applications*. Boca Raton, FL, USA: CRC Press, 2025.

[28] N. K. Mishra and P. K. Singh, "Feature construction and smote-based imbalance handling for multi-label learning," *Inf. Sci.*, vol. 563, pp. 342–357, Jul. 2021, doi: 10.1016/j.ins.2021.03.001.

[29] F. Cady, *The Data Science Handbook*. Hoboken, NJ, USA: Wiley, 2024.

[30] B. K. Iwana and S. Uchida, "An empirical survey of data augmentation for time series classification with neural networks," *PLoS ONE*, vol. 16, no. 7, Jul. 2021, Art. no. e0254841, doi: 10.1371/journal.pone.0254841.

[31] P. F. Suawa, A. Halbinger, M. Jongmanns, and M. Reichenbach, "Noise-robust machine learning models for predictive maintenance applications," *IEEE Sensors J.*, vol. 23, no. 13, pp. 15081–15092, Jul. 2023, doi: 10.1109/JSEN.2023.3273458.

[32] *Pattern Recognition and Machine Learning | SpringerLink*. Accessed: Dec. 1, 2024. [Online]. Available: https://link.springer.com/book/9780387310732

[33] L. B. V. De Amorim, G. D. C. Cavalcanti, and R. M. O. Cruz, "The choice of scaling technique matters for classification performance," *Appl. Soft Comput.*, vol. 133, Jan. 2023, Art. no. 109924, doi: 10.1016/j.asoc.2022.109924.

[34] A. Shmilovici, "Support vector machines," in *Machine Learning for Data Science Handbook: Data Mining and Knowledge Discovery Handbook*. Cham, Switzerland: Springer, 2023, pp. 93–110, doi: 10.1007/978-3-031-24628-9_6.

[35] J. C. Y. Ngu, W. S. Yeo, T. F. Thien, and J. Nandong, "A comprehensive overview of the applications of kernel functions and data-driven models in regression and classification tasks in the context of software sensors," *Appl. Soft Comput.*, vol. 164, Oct. 2024, Art. no. 111975, doi: 10.1016/j.asoc.2024.111975.

[36] J. Wainer and P. Fonseca, "How to tune the RBF SVM hyperparameters? An empirical evaluation of 18 search algorithms," *Artif. Intell. Rev.*, vol. 54, no. 6, pp. 4771–4797, Aug. 2021, doi: 10.1007/s10462-021-10011-5.

[37] D. M. Dziuda, Ed., "Regression with random forests," in *Multivariate Biomarker Discovery: Data Science Methods for Efficient Analysis of High-Dimensional Biomedical Data*. Cambridge, U.K.: Cambridge Univ. Press, 2024, pp. 126–135, doi: 10.1017/9781009006767.011.

[38] H. L. Smith, P. J. Biggs, N. P. French, A. N. H. Smith, and J. C. Marshall, "Out of (the) bag—Encoding categorical predictors impacts out-of-bag samples," *PeerJ Comput. Sci.*, vol. 10, p. e2445, Nov. 2024, doi: 10.7717/peerj-cs.2445.

[39] B. Liu and R. Mazumder, "Randomization can reduce both bias and variance: A case study in random forests," 2024, *arXiv:2402.12668*.

[40] *XGBoost: The Ultimate Guide to Extreme Gradient Boosting EBook: Van Maarseveen, Henri: Amazon.ca: Books*. Accessed: Dec. 1, 2024. [Online]. Available: https://www.amazon.ca/XGBoost-Ultimate-Extreme-Gradient-Boosting-ebook/dp/B0C5JGW1SR

[41] Q. Wang, X. Zou, Y. Chen, Z. Zhu, C. Yan, P. Shan, S. Wang, and Y. Fu, "XGBoost algorithm assisted multi-component quantitative analysis with Raman spectroscopy," *Spectrochimica Acta A, Mol. Biomolecular Spectrosc.*, vol. 323, Dec. 2024, Art. no. 124917, doi: 10.1016/j.saa.2024.124917.

[42] I. H. Witten, E. Frank, M. A. Hall, C. J. Pal, and J. Foulds, *Data Mining: Practical Machine Learning Tools and Techniques*. Amsterdam, The Netherlands: Elsevier, 2025.

[43] M. Zuo and Y. Liu, "Convolutional neural network modulation recognition by incorporating attention mechanisms," in *Proc. 3rd Int. Conf. Intell. Commun. Comput. (ICC)*, Nov. 2023, pp. 335–339, doi: 10.1109/ICC59986.2023.10421616.

[44] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016. [Online]. Available: http://www.deeplearningbook.org

[45] Z. Niu, G. Zhong, and H. Yu, "A review on the attention mechanism of deep learning," *Neurocomputing*, vol. 452, pp. 48–62, Sep. 2021, doi: 10.1016/j.neucom.2021.03.091.

[46] C. Wei, Z. Quan, Z. Qian, H. Pang, Y. Su, and L. Wang, "An attention mechanism augmented CNN-GRU method integrating optimized variational mode decomposition and frequency feature classification for complex signal forecasting," *Expert Syst. Appl.*, vol. 269, Apr. 2025, Art. no. 126464, doi: 10.1016/j.eswa.2025.126464.

[47] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2011, pp. 1–7. Accessed: Dec. 1, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2011/hash/86e8f7ab32cfd12577bc2619bc635690-Abstract.html

[48] Z.-H. Zhou, *Ensemble Methods: Foundations and Algorithms*. Boca Raton, FL, USA: CRC Press, 2025.

[49] T. O. Hodson, "Root-mean-square error (RMSE) or mean absolute error (MAE): When to use them or not," *Geosci. Model Develop.*, vol. 15, no. 14, pp. 5481–5487, Jul. 2022, doi: 10.5194/gmd-15-5481-2022.

[50] P. Agrawal, C. Gupta, A. Sharma, V. Madaan, and N. Joshi, Eds., *Machine Learning and Data Science: Fundamentals and Applications*, 1st ed., Hoboken, NJ, USA: Wiley, 2022.

[51] G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, *An Introduction to Statistical Learning: With Applications in Python*. Cham, Switzerland: Springer, 2023.

[52] S. Raza, "A machine learning model for predicting, diagnosing, and mitigating health disparities in hospital readmission," *Healthcare Analytics*, vol. 2, Nov. 2022, Art. no. 100100, doi: 10.1016/j.health.2022.100100.

[53] M. A. Mansournia, R. Waters, M. Nazemipour, M. Bland, and D. G. Altman, "Bland–Altman methods for comparing methods of measurement and response to criticisms," *Global Epidemiol.*, vol. 3, Nov. 2021, Art. no. 100045, doi: 10.1016/j.gloepi.2020.100045.

[54] M. Zeynali, K. Alipour, B. Tarvirdizadeh, and M. Ghamari, "Non-invasive blood glucose monitoring using PPG signals with various deep learning models and implementation using TinyML," *Sci. Rep.*, vol. 15, no. 1, p. 581, Jan. 2025, doi: 10.1038/s41598-024-84265-8.

[55] Y. Sun and P. Kosmas, "Noninvasive glucose prediction system enhanced by mixed linear models and meta-forests for domain generalization," *IEEE Sensors J.*, vol. 25, no. 8, pp. 14209–14219, Apr. 2025, doi: 10.1109/JSEN.2025.3542385.

[56] M. Tan, M. A. Merrill, V. Gupta, T. Althoff, and T. Hartvigsen, "Are language models actually useful for time series forecasting?" 2024, *arXiv:2406.16964*.

[57] Q. Zhang, Q. Liu, and Q. Ye, "An attention-based temporal convolutional network method for predicting remaining useful life of aero-engine," *Eng. Appl. Artif. Intell.*, vol. 127, Jan. 2024, Art. no. 107241, doi: 10.1016/j.engappai.2023.107241.

[58] Q. Fournier, G. M. Caron, and D. Aloise, "A practical survey on faster and lighter transformers," *ACM Comput. Surveys*, vol. 55, no. 14s, pp. 1–40, Jul. 2023, doi: 10.1145/3586074.

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